

History, Version and features of Python:

1. History of Python :-

- Python was created by Guido van Rossum.
- He started working on Python in December 1989 at CWI (Centrum Wiskunde & Informatica) in the Netherlands.
- Python was designed to be a simple, readable and powerful programming language.
- It was first released in February 1991.
- The name "Python" comes from the British comedy group "Monty Python's Flying Circus", which Guido van Rossum was a fan of.

2. Versions of Python :-

Version	Release Year	Key Highlights
Python 1.0	1994	First official release with basic features.
Python 1.5	1997	Added modules, exception handling.
Python 2.0	2000	List comprehensions, Garbage collection improved.
Python 2.7	2010	Last release of Python 2.x series.
Python 3.0	2008	Major update, not backward compatible.
Python 3.x (latest)	2024	Better performance, new features and improvements.

3. Features of Python :-

- ① Simple and Easy to Learn : Python has simple syntax, similar to English.
- ② Readable Code : Its code is clean and easy to read.
- ③ Interpreted Language : Python does not need compilation. It runs line by line.
- ④ High-Level Language : It abstracts low-level details.
- ⑤ Portable : Python code can run on Windows, Linux, macOS without changes.
- ⑥ Object-Oriented : Supports classes and objects.
- ⑦ Rich Standard Library : Comes with a large collection of built-in modules.
- ⑧ Extensible : Can be extended using C, C++ and Java.
- ⑨ Dynamically Typed : No need to declare variable types.
- ⑩ Versatile : Used in Web Development, Data Science, AI/ML, Automation, etc.

Working with Python - Writing and running Code.

Python code is written in a file with .py extension and then executed (run) using the Python interpreter.

1. Writing Python Code

- Open a code editor like . IDLE, VS Code, PyCharm, or any text editor.
- Create a new file.
- Write your Python code.
- Save the file with .py extension. (example: `hello.py`)

2. Example - Simple Python Program

<code># This is my first Python program</code>	<code># single line comment</code>
<code>print ("Hello, World!")</code>	<code># print statement</code>
<code>name = "Python"</code>	<code># variable declaration</code>
<code>print ("Welcome to", name)</code>	<code># output with variable</code>

3. Running Python Code

A. Using IDLE.

- ① Open IDLE.
- ② Click on File → Open and select your .py file.
- ③ Click on Run → Run Module (or press F5).
- ④ Output will appear in the Shell window.

B. Using Command Prompt / Terminal

- ① Open Command Prompt (cmd) or Terminal.
- ② Navigate to the folder where your .py file is saved.
Use `cd` command.
- ③ Run the program using : `python filename.py`
Example : `python hello.py`
- ④ Output will be displayed in the terminal.

4. Output

```
Hello, World!  
Welcome to Python
```

Make sure Python is installed
and added to PATH.

Note : Always save your file before running the code.



Python installation and Environment Setup.

To start programming in Python, we first need to install Python and set up the development environment.

1. Python Installation (Windows)

- ① Go to the official Python website : <https://www.python.org>
- ② Click on Downloads → Choose the latest Python 3.x version.
- ③ Click the installer (Windows installer).
- ④ Run the downloaded file.
- ⑤ **Important:** Check the box "Add Python to PATH" before clicking "Install Now".
- ⑥ Wait for installation to complete.
- ⑦ Click "Close" after successful installation.

2. Verify Installation

- Open Command Prompt (cmd) and type :

```
python --version
```

- You should see the installed Python version, e.g.

Python 3.12.3

3. Environment Setup

- There are different ways to set up Python environment.

A. Using IDLE (comes with Python)

- IDLE is a simple IDE for Python.
- Open IDLE from Start Menu → Python 3.x → IDLE (64-bit).
- You can start writing and running Python code.

B. Using VS Code (Recommended)

- Download and install Visual Studio Code from <https://code.visualstudio.com>
- Install the "Python" extension from Extensions Marketplace.
- Open a folder → Create a .py file → Write and run code.

4. Virtual Environment (Recommended for Projects)

- A virtual environment keeps project dependencies separate.

• Steps :

- ① Open Command Prompt in your project folder.
- ② Create virtual environment : `python -m venv myenv`
- ③ Activate environment : `myenv\Scripts\activate`
- ④ Your prompt will change, now you are inside the virtual env.
- ⑤ Install packages using pip, e.g., `pip install requests`
- ⑥ Deactivate when done : `deactivate`

Note : Always work inside a virtual environment for your projects.



Features of Python

Python is a powerful, high-level programming language. It has many useful features that make it popular and easy to use.

- ① **Easy to Learn and Use** : Python has simple syntax that is easy to understand, even for beginners.
- ② **Readable and Clean Code** : Python code is clean and easy to read. It uses indentation to define code blocks.
- ③ **Interpreted Language** : Python is an interpreted language, which means code is executed line by line. There is no need for compilation.
- ④ **High-Level Language** : Python abstracts low-level details (like memory management), so we can focus on solving problems.
- ⑤ **Dynamically Typed** : Python does not require declaration of variable types. Types are decided at runtime automatically.
- ⑥ **Object-Oriented** : Python supports object-oriented programming. It supports classes, objects, inheritance, polymorphism, etc.
- ⑦ **Portable** : Python is platform independent. Python programs can run on Windows, Linux, macOS, and other platforms without major changes.
- ⑧ **Rich Standard Library** : Python comes with a large collection of built-in modules and functions. We can do a lot without installing anything.
- ⑨ **Extensible** : Python can be extended using C, C++ or other languages. We can also create our own modules.
- ⑩ **Versatile / Multi-purpose** : Python can be used in web development, data science, AI/ML, automation, scientific computing, game development, and more.
- ⑪ **Large Community Support** : Python has a huge community. We get many resources, tutorials, and third-party libraries.

Version of Python.

Python has had several major versions over the years. Here are the important versions and their key highlights.

<u>Version</u>	<u>Release Year</u>	<u>Key Highlights</u>
1. Python 1.0	1994	• First official release by Guido van Rossum. • Included basic features like functions, modules, exceptions, and core data types.
2. Python 1.5	1997	• Added support for Unicode. • Introduced built-in functions like <code>any()</code> and <code>all()</code>.
3. Python 2.0	2000	• Major new release with many new features. • Introduced list comprehensions and garbage collection. • Better memory management.
4. Python 2.7	2010	• Last and most stable release of Python 2.x. • Extensive standard library and wide adoption. • Official support ended in 2020.
5. Python 3.0	2008	• Major update, not backward compatible. • Introduced <code>print()</code> function. • Better Unicode support. • Division operator <code>/</code> gives float result.
6. Python 3.1 - 3.9	2009 - 2020	• Many new features and optimizations. • Example: type hints, f-strings, new standard library modules, performance improvements.
7. Python 3.10+	2021 onwards	• Pattern matching (<code>match-case</code>). • Better error messages. • Continued performance improvements and new features.

Note : Python 3.x is the latest and recommended version for all new projects.



History of Python

- Python was created by Guido van Rossum.
- He started working on Python in December 1989 at CWI (Centrum Wiskunde & Informatica) in the Netherlands.
- The language was designed with a goal of writing simple, readable and powerful code.
- Python was influenced by the programming language ABC, which was also created by Guido van Rossum.
- It was first released in February 1991.
- The name "Python" comes from the British comedy group "Monty Python's Flying Circus", which Guido van Rossum was a fan of.
- Python 1.0 was released in 1994 with basic features such as functions, modules, exceptions and data types.
- Over the years, Python has evolved and new versions have added many powerful features and improvements.
- Python 2.0 was released in 2000.
- Python 3.0 was released in 2008 with major changes to make the language more advanced and consistent.
- Python continues to grow and is now one of the most popular programming languages in the world.